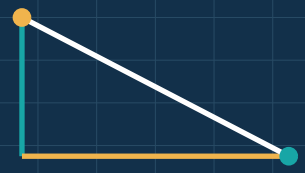
ENGINEERING GRAPHICS

AI PERSONAL STUDY ASSISTANT



PROJECT REPORT

PROJECTIONS • ROTATION • TRACES

STUDENT	NEMAY R
DEPARTMENT	Artificial Intelligence & Machine Learning (AIML)
COLLEGE	TKMIT
ACADEMIC YEAR	2026–2030
SYLLABUS	KTU 2024 scheme — confirm exact course outline with faculty

ABOUT THIS REPORT
An illustrated quick-revision guide to Engineering Graphics, with key construction steps, diagrams, and 18 practice questions. Use your official syllabus and class notes for exam-specific requirements.

LEARN • DRAW • CHECK • IMPROVE

01 | Fundamentals & Standards

Engineering Graphics communicates the shape, size, position, and details of objects through standard technical drawings. Good work is accurate, legible, consistent, and properly dimensioned.

Learning goals

- Read standard line conventions and dimensions.
- Construct projections of points, lines, planes, and solids.
- Understand rotation, traces, sections, development, and pictorial views.

Instruments & line conventions

Line	Common use
Continuous thick	Visible outlines and edges
Continuous thin	Dimensions, extension and projection lines
Dashed thin	Hidden edges
Chain thin	Centre lines and axes

DRAFTING TIP
Keep construction lines light, final outlines dark, lettering consistent, and dimensions clear. Avoid repeating the same dimension.

Quick check

1. Name two instruments used for circles/arcs. 2. Which line type is used for hidden edges?

02 | Projection of Points & Lines

Orthographic projection represents a 3D object on principal reference planes. HP is the horizontal plane, VP is the vertical plane, and XY is their intersection. In first-angle projection, follow the view placement taught in your course.

Projection of a point

The front view shows the point's height relative to HP; the top view shows its distance from VP. Use projectors perpendicular to XY so corresponding views remain aligned.

Projection of a straight line

A line parallel to a projection plane appears in true length on that plane. An inclined line generally appears foreshortened. Label corresponding endpoints consistently (A, B, a, b, a′, b′).

> **CONSTRUCTION SEQUENCE**
> Draw XY → mark given distances → project perpendicular to XY → locate both views → label and check alignment.

Practice questions

1. Point A is 25 mm above HP and 30 mm in front of VP. Draw its front and top views.

2. A 70 mm line is parallel to VP and inclined at 30° to HP. Which view shows its true length? Explain briefly.

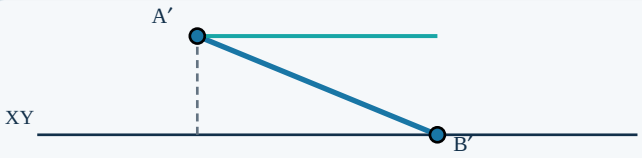
3. Explain the difference between true length and apparent length.

03 | Rotation of Lines & Planes

The rotation method changes the position of a view around a fixed point or axis to satisfy a given inclination. Use it to establish the required projection while preserving the appropriate distances.

Rotation of a line

1. Draw initial projections. 2. Keep the specified endpoint fixed. 3. Rotate the relevant view through the given angle. 4. Use an arc to preserve distance. 5. Project the rotated endpoint to obtain the corresponding view.



Rotate about the fixed point.

Preserve true length using an arc.

Rotation of a plane

For a lamina, establish a convenient edge/reference position, rotate it to the required inclination, project all corners, and join them in sequence. Keep vertex labels consistent.

Common pitfalls

- Rotating about the wrong point or axis.
- Changing the true length during rotation.
- Confusing angles to HP with angles to VP.

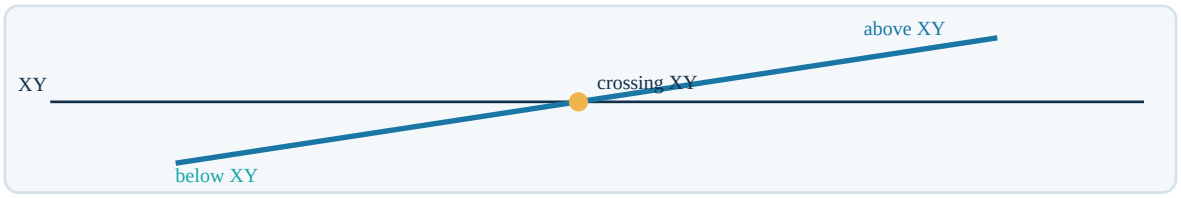
Practice questions

4. A 65 mm line has one end 15 mm above HP and 20 mm in front of VP. It is inclined at 35° to HP and 45° to VP. Outline the rotation-method steps.

5. Describe the construction sequence for a rectangular lamina inclined to HP, with one edge parallel to VP.

04 | Traces of Lines & Planes

A trace is where an extended line or plane meets a principal reference plane. HT (horizontal trace) lies on HP; VT (vertical trace) lies on VP.



Finding traces of a straight line

- Extend the front view to meet XY; project that location to the top view to find HT.
- Extend the top view to meet XY; project that location to the front view to find VT.
- Label HT and VT; use light construction projectors.

Traces of a plane

A plane trace is the line where the plane intersects HP or VP. Establish the plane orientation and use its edges or a suitable construction to locate the intersection line.

> **REMEMBER**
> A line parallel to HP has no finite HT; a line parallel to VP has no finite VT.

Practice questions

6. Define HT and VT. Explain how each is located from front and top views.

7. Draw a schematic line crossing XY and label the construction needed to locate its traces.

8. What is the difference between the trace of a line and the trace of a plane?

05 | Solids, Sections & Development

Solids

For prisms, pyramids, cylinders, and cones, begin with the view that best shows the base. Project the axis and characteristic edges into the other view, keeping corresponding features aligned.

Sections

A section is formed when an imaginary cutting plane passes through a solid. Locate the intersections with edges/surfaces, project the points into the required view, join them in sequence, and hatch the cut area with thin, evenly spaced lines.

Development

Development unfolds a solid's surface onto a plane. Preserve true lengths and correct spacing; mark the seam clearly.

Solid	Method	Check
Prism / cylinder	Parallel-line	Equal spacing / circumference
Pyramid / cone	Radial-line	True slant edges / height

Practice questions

9. A hexagonal prism rests on HP on its base. Describe the views needed when its axis is perpendicular to HP.

10. Explain the steps for drawing the sectional front view of a solid cut by an inclined plane.

11. Which development method is used for a cone? What dimension must be preserved?

12. What is the purpose of hatching in a sectional view?

06 | Orthographic, Isometric & Perspective

Orthographic views

Front, top, and side views show shape and dimensions accurately. Select a useful front view, align the views with projectors, and use hidden lines where required.

Isometric drawing

Three principal directions are represented by axes 120° apart. Isometric drawings usually show three faces. In an isometric view, lengths along axes are commonly drawn at true scale; strict isometric projection uses an isometric scale. Circles appear as ellipses.

Perspective drawing

Perspective aims for realistic appearance. Receding parallel edges may converge towards one or more vanishing points on the horizon line.

View type	Purpose	Key feature
Orthographic	Accurate dimensions	Several aligned views
Isometric	3D pictorial view	Three visible directions
Perspective	Visual realism	Vanishing point(s)

Practice questions

13. Draw front, top, and right-side views of a simple stepped block, keeping edges aligned.

14. Sketch a cube in isometric view and label the three axes.

15. Give two differences between orthographic and perspective drawings.

07 | Practice Set & Revision Plan

HOW TO USE THE PRACTICE SET
Attempt each question on paper first. Show construction lines, use clear labels, and check dimensions before comparing with notes.

Final practice questions

16. List four engineering drawing line types and give one use for each.

17. A point is 35 mm below HP and 25 mm behind VP. Identify its quadrant and describe the view positions using your course's first-angle convention.

18. A cylinder is cut by a plane parallel to its base. State the shape of the section and explain how it is shown.

Selected self-check answers

1. Front view 25 mm above XY; top view 30 mm below XY under the usual first-angle setup. 2. Its front view shows true length when the line is parallel to VP. 3. True length is actual length; apparent length is a foreshortened projection. 11. Radial-line method; preserve true slant height. 12. Hatching identifies cut material. 15. Orthographic uses multiple views for dimension accuracy; perspective depicts visual depth. 17. Third quadrant; check the exact prescribed view placement. 18. A circle for a right circular cylinder cut parallel to its base.

7-day revision plan

Day 1: Standards & points • **Day 2:** Lines • **Day 3:** Rotation • **Day 4:** Traces • **Day 5:** Solids & sections • **Day 6:** Development & pictorial views • **Day 7:** Timed practice + corrections

CONCLUSION
Engineering Graphics builds spatial reasoning and technical communication. Practise constructions in stages, keep drawings neat, and verify topics against the official KTU 2024 syllabus and faculty notes.

Reference note: Official KTU syllabus, prescribed textbook, and faculty instructions take precedence for exam-specific content and conventions.